Research paper analysis: K. Geevers et al.

"Policy based PPO"

- multi-echelon inventory optimization problem with general network structure.
- Stock points - (a) warehouse (b) retailers
- Markov Decision Process (MDP)

  (time: t, state: s_t, action: a_t)

* Time period is <u>NOT</u> a state variable. Looking for a stationary policy that is valid at any time period.

- Action space: order for every edge of the network.
- $S_t \rightarrow S_{t+1}$ depends on action space & given by $P(S_{t+1} | S_t, a_t)$

- 5 events happen-

(1) Shipment of upstream stock points received

(2) Supplier sends outgoing shipment to warehouse with stochastic lead time λ_w

(3) Retailer fulfills orders from retailers with on-hand inventory.

Transport stochastic time- λ_{wr}

(4) If not enough inventory:

rank (retailer) in asc. order of $\underbrace{\text{inventory} - \text{backorder}}_{(A)}$

lowest (A) satisfied first until no inventory left

when no inventory left, partial order goes to backorder

After sending outgoing shipment, upstream orders are placed

(5) Finally cust. demand fulfilled at retailer & demand not satisfied is placed in backorder

* Demand of cust. is not known before replenishment orders are placed

Cost fn. is eq (3); aim is to minimise

$$c_t(s_t, a_t) = \sum_{w \in W} \left[h_w \cdot I_{w,t} + b_w \cdot \sum_{r \in R} BO_{r,w,t} \right] + \sum_{r \in R} \left[h_r \cdot I_{r,t} + b_r \cdot BO_{r,t} \right] \quad (3)$$

The objective of the MDP is to find a policy π^* that minimizes the discounted infinite sum of costs, where a policy is a mapping of states to actions. Thus, the

Objective of MDP: find policy π^* that minimizes discounted infinite sum of costs as expressed in eq. (4)

Implementation of PPO: 2 NN approximators
one for policy fn. (actor) & one for value fn. (critic) which do not share weight.

Actor: given i/p state, get action taken in the state
Critic: outputs estimated value of being in that state
Network weights initialised randomly to encourage exploration at beginning of training
* O/p of actor network results in stochastic property, as o/p consists of mean & s.d. of Gaussian distribution

- In every iteration, buffer is filled with states, actions, reward following current policy.
- Advantage calculated using Generalised Advantage Estimation
- Based on advantage, paras. of both NN are updated by loss fn. based on trajectory of current policy.
  ↳ clipped surrogate objective
- Learning continues until some convergence is reached

Hyper parameters in Table 4

- Normalise i/p to [-1, 1] to avoid vanishing or exploding gradient

$$\pi^* = \arg\min_{\pi \in \Pi} \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t \cdot c_t\left(s_t, a_t^{\pi}\right)\right] \tag{4}$$

Table 4 PPO hyper-parameters

Hyper-parameter	Parameter value
# of iterations	10,000 (Linear, Divergent); 50,000 (general)
Layers	2 hidden layers with 64 nodes each: (64, 64)
Activation function	tanh
Initialization of weights	Glorot-Uniform
Buffer size	256
Discount factor (γ)	0.99
GAE parameters (λ)	0.95
ϵ	0.2
Adam learning rate	1e−4
Batch size	64
Update epochs	10

Action space - ~~discrete~~ continuous

no. (o/p nodes) = 2 x no. (diff. comb. of variable values)

Implementation: order per edge

Scale action space to [-1, 1]

O/p is unbounded, clip o/p that lie outside given interval

Rewards - MAXIMIZE

cost mult. by -1 gives rewards

- Since using Actor-Critic, try to achieve minimal MSE loss
- Dividing rewards by 1000 yielded good results

Environment - DIVERGENT

- described by Kunnumkai & Topaloglu
- Inventory Distribution System consisting of single warehouse & multiple retailers

(a) retailers face random demand from end customers

(b) unsatisfied demand at retailers is backlogged

(c) warehouse supplies retailers & replinishes inventory from external supplier

(d) inventory can be stocked at warehouse & any retailers

- lead time is deterministic
- demand follows Poisson distributions (mean from uniform distribution)
- no backlogged at warehouse
- four stock points

Algo. parameters for divergent only from Table 5

Experiment Setup -

- Iterations : 10K
- Simulate learned policy every 100 iterations
- Intel Xeon Processor CPU @ 2.20 GHz with 4 GB RAM
- One full replication takes 1 hr

Upper bounds for divergent only from Table 6

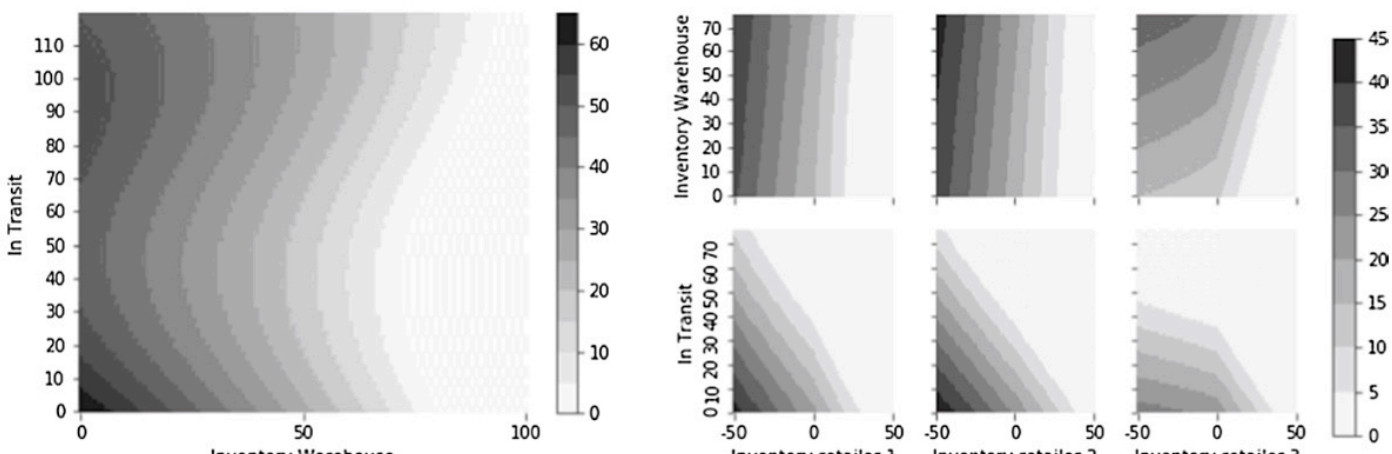
Heat Map from fig. 4

Table 5 Parameters per case

Structure	λ_w	$\lambda_{w,r}$	d_i	K	h_w	h_r	b_w	b_r
Linear	Unif{0, 4}	Unif{0, 4}	Unif{0, 15}	4	1	1	2	2
Divergent	1	1	Pois(Unif{5, 15})	4	1	1	0	19
General	1	1	Pois(15)	9	0.6	1	0	19

Table 6 Upper bound of the state and action variables per case

Structure	ti	tb	$I^{end}_{w,t}$	$I^{end}_{r,t}$	$BO_{r,w,t}$	$BO_{r,t}$	$IT_{w,t,t'}$	$IT_{w,r,t,t'}$	$O_{w,t}$	$O_{r,w,t}$
Linear	4000	4000	1000	1000	1000	1000	150	150	30	30
Divergent	1000	450	250	250	150	150	300	75	300	75
General	4500	8250	500	500	500	250	150	75	150	75



(a) Warehouse. (b) Retailers.

Fig. 4 Actions of the warehouse and retailers